

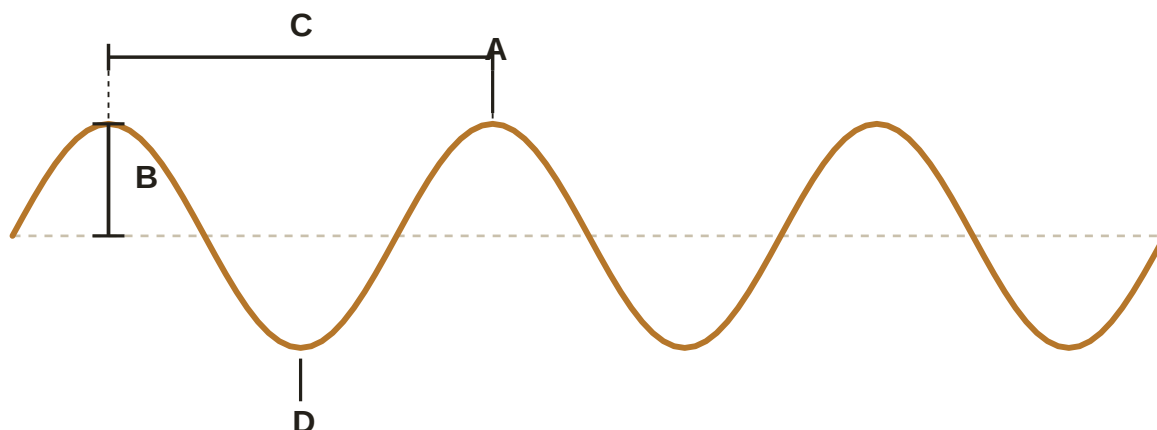
Wave Basics + Standard Form

Test · 26 marks total · no calculator needed for standard form questions

Name: _____ Date: _____ Score: _____ / 26

Section A — Label the diagram

The diagram below has four labelled points, A–D. Identify what each one shows.



A: _____

B: _____

C: _____

D: _____

[4 marks]

Section B — Wave concepts

B1. Describe, in terms of particle movement, what makes a wave transverse.

[2 marks]

B2. Give one example of a longitudinal wave.

[1 mark]

B3. A wave has a frequency of 25 Hz and a wavelength of 12 m. Calculate its speed. State the equation you use.

[3 marks]

B4. A wave travels at 320 m/s and has a frequency of 40 Hz. Calculate its wavelength.

[3 marks]

B5. Two waves have the same wavelength but different amplitudes. Which carries more energy, and why?

[2 marks]

Section C — Standard form practice

C1. Write 6200 in standard form. [1 mark]

C2. Write 0.0034 in standard form. [1 mark]

C3. Write 2.5×10^4 as an ordinary number. [1 mark]

C4. Write 7×10^{-3} as an ordinary number. [1 mark]

C5. A gamma ray has a wavelength of 0.00000000001 m. Write this in standard form. [1 mark]

C6. A long-wave radio signal has a wavelength of 1200 m. Write this in standard form. [1 mark]

C7. The speed of light is 300 000 000 m/s. Write this in standard form. [1 mark]

C8. A wave's time period is 0.00025 s. Write this in standard form. [1 mark]

C9. Write 5.6×10^{-5} as an ordinary decimal number. [1 mark]

C10. Put these in order, smallest to largest: 3×10^{-2} , 4.5×10^{-4} , 2×10^{-1} [2 marks]
